Harsh Shelke

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Education

University of Illinois at Chicago — Chicago,IL
Masters of Science in Computer Science — *Aug. 2024 – May 2026*

MIT World Peace University — Pune,INDIA
Bachelor of Technology-Computer Science and Engineering — *Aug. 2020 – June 2024*

Technical Skills

Languages: Java, Python, C, C++, SQL, JavaScript, TypeScript, Dart, Bash, Shell, MATLAB, HTML, CSS
Frameworks: Angular, Vue.js, Next.js, Node.js, Flask, FastAPI, Django, ASP.NET, Material-UI, REST API, TensorFlow, PyTorch, Keras, LangChain, Hugging Face Transformers
Developer Tools: Git, Docker, Travis CI, Google Cloud Platform (GCP), VS Code, Visual Studio, PyCharm, Cursor
Libraries: React, NumPy, Pandas, Matplotlib, SciPy (scipy.stats), Scikit-learn, XGBoost

Experience

Software Intern — Jan 2024 – Jun 2024
BizAmica Pvt.Ltd. — *Pune, INDIA*

- Engineered document ingestion pipelines with **Flask** to automate manual data entry, file parsing, and validation workflows, handling **over 50,000 records daily** with consistent accuracy and faster turnaround.
- Developed airway bill and tax processing pipelines using **Flask, Pandas, and spaCy**, designing prompts with **OpenAI APIs** for automated data extraction that boosted compliance verification accuracy to **93%**.
- Designed **Angular** dashboards to track client invoice sales in real time, enabling **200+** active users to streamline financial monitoring and reduce manual oversight.

ML Intern — Aug 2023 – Jan 2024
Al Adventures LLM — *Pune, INDIA*

- Programmed customer lifetime value prediction models using **TabNet, XGBoost, and TensorFlow**, lowering mean absolute error by **13%** to support more accurate prioritization of high-value customer opportunities.
- Devised advanced hypothetical **z, t, A/B, and chi-squared tests** using **SciPy, Statsmodels, and PyMC3** on transaction data, identifying **2,300** high-risk customers for retention campaigns.
- Developed exploratory data analysis and visualizations with **Plotly, Seaborn, and Tableau** on phishing website data, detecting URL obfuscation that helped peers shorten incident response time by **20 minutes**.

Projects

ML-enabled Moments | *Flask,Hugging Face,Vision Transformer(ViT)+GPT-2,Microsoft Azure* — September 2025

- Architected ML pipeline using **Hugging Face Transformers (ViT-GPT2)** for automated image captioning, leveraging 1-2GB pre-trained models and reducing manual alt-text generation by **80%.**
- Integrated **Azure Computer Vision API** for automated object detection and tagging, enabling search across many object types with **89%** accuracy.
- Constructed Trending Tags and Latest AI Tag functionality to enhance photo discovery using **Flask, SQLAlchemy, indexed SQL, and YOLOv8** which currently support **500+ tags.**

NLP Resume Parser | *Python, SpaCy, Streamlit, Hugging Face, Pandas, NumPy, PDFPlumber* — May 2025

- Developed an NLP pipeline with **SpaCy, PDFPlumber, KeyBERT, and Sentence Transformers** to extract text, skills, degrees, and organizations from PDFs, reducing recruiter's review time by **70%**.
- Containerized a **Streamlit application** and deployed on **Hugging Face Spaces** for on-demand resume processing, enabling reliable real-time parsing throughput of **500+ resumes per day.**
- Automated skills and role matching using **embeddings-based cosine similarity and KeyBERT** to surface ranked insights,extracting the **top 15 skills** per resume to guide recruiter's decisions.

HarshFit-Fitness Application | *React, Tailwind CSS, Netlify, JavaScript* — December 2024

- Spearheaded a **React-based fitness app** with workout cards and modal components using **JavaScript and Tailwind CSS**, reducing workout planning time from **20 minutes to under 3 minutes per session**.
- Orchestrated a **JavaScript** recommendation system that filtered exercises by muscle groups and training goals using **React hooks**, generating personalized workouts with **95%+ accuracy** across 4 training splits.
- Implemented a client-only deployment pipeline using **React, Vite, and Tailwind CSS on Netlify** with continuous deployment, delivering **0 server-side endpoints** to simplify operations and maintenance.